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Variable speed limit control for severe non-recurrent freeway bottlenecks

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ABSTRACT

Variable speed limit (VSL) schemes are developed based on the Kinematic Wave theory to increase discharge rates at severe freeway bottlenecks induced by non-recurrent road events such as incidents or work zones while smoothing speed transition. The main control principle is to restrict upstream demand (in free-flow) progressively to achieve three important objectives: (i) to provide gradual speed transition at the tail of an event-induced queue, (ii) to clear the queue around the bottleneck, and (iii) to discharge traffic at the stable maximum flow that can be sustained at the bottleneck without breakdown. These control objectives are accomplished without imposing overly restrictive speed limits. We further provide remedies for (a) underutilized bottleneck capacity due to underestimated stable maximum flow and (b) a re-emergent queue at the bottleneck due to an overestimated stable maximum flow. We analytically formulate the reductions in total delay in terms of control parameters to provide an insight into the system performance and sensitivity. The results from the parameter analysis suggest that significant delay savings can be realized with the proposed VSL control strategies.

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1. Introduction

Variable speed limit (VSL) control seeks to improve safety by smoothing out traffic state transitions temporally and/or spatially and freeway efficiency by deferring the onset of congestion or increasing the bottleneck (BN) discharge rate. Earlier efforts to improve freeway efficiency via VSL focused on harmonizing the speed across vehicles in different lanes to create a more homogenous, stable one-pipe flow with few lane-changes (LC) (Smulders, 1990; Zackor, 1991). This can presumably lead to higher capacity and critical density, thereby deferring or preventing onset of congestion (Papageorgiou et al., 2008). Several studies have shown that VSL control indeed induces more balanced speed and utilization of lanes (Duret et al., 2012; Knoop et al., 2010; Smulders, 1990; Weikl et al., 2013; Zackor, 1991).

SPECIALIST (SPEed ControllIng ALgorIthm using Shock wave Theory) seeks to proactively resolve a moving jam and maximize the discharge rate by limiting the speed and density of the inflow to the moving jam via VSL (Hegyi et al., 2008, 2009). The algorithm was tested in the field with reasonable success. Another notable scheme is the mainstream traffic flow control (MTFC) developed in the framework of discrete-time optimal control (Carlson et al., 2010a). The main objective is to control the free-flow traffic upstream of a BN (before a queue arises) via VSL or ramp metering to prevent BN activation. The algorithm was tested on a Dutch network via simulation (Carlson et al., 2010b). Later, local feedback control was incorporated into MTFC to further improve the potential for field implementation (Carlson et al., 2011).

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Chen et al. (2014) developed different VSL schemes based on the Kinematic Wave (KW) theory (Lighthill and Whitham, 1955; Richards, 1956) to increase freeway BN discharge rates and manage the queue upstream (for smoother speed transition) under two scenarios: steady queue and oscillatory queue that can inevitably arise at fixed BNs. The key principle is to impose VSL control some distance upstream of a BN to starve the inflow to the BN and dissipate the queue. Once the queue near the BN vanishes, another less restrictive VSL is imposed upstream to (i) resolve the heavy queue generated by the first VSL and (ii) regulate the inflow to sustain the stable maximum BN discharge rate and prevent BN re-activation.

The strategies cited above were designed primarily to address recurrent BNs or moving jams, in which a reduction in discharge rate typically ranges from 5% to 15% (e.g., Bertini and Leal, 2005; Cassidy and Bertini, 1999). These strategies, however, are not very suitable for non-recurrent severe BNs induced by major road events such as incidents or work zones. These types of severe BNs usually involve (partial) lane blockage and are characterized by moderate to severe congestion due to significant reductions in system throughput (in greater proportion than lanes blocked (Knoop et al., 2008, 2009a,b)) and sharp transitions upstream from free-flow traffic to the queues, which may cause secondary incidents. A significant reduction in system throughput is attributable to: (1) a decrease in capacity due to road blockage, (2) rubbernecking around the road event, (3) change in driver characteristics (Knoop et al., 2009b), and (4) disruptive LCs away from the road event. In regard to (4), speed and flow may vary significantly among lanes with lower speed and flow closer to the road event. LCs away from the road event are likely to create voids in other lanes and reduce the discharge rate similar to the capacity drop phenomenon of recurrent BNs (Laval and Daganzo, 2006).

In this study, we develop VSL strategies to proactively improve the discharge rate of non-recurrent severe BNs and manage the upstream queue for smoother speed transition. An increase in discharge rate is achieved by clearing the queue around the BN and then maintaining a stable maximum flow with harmonized speed to minimize disruptive LCs. We assume that other aforementioned effects on discharge rate reductions ((1)–(3) in the previous paragraph) are reflected in the FD of the BN (e.g., rubbernecking may decrease the free-flow speed), but our VSL strategies do not significantly change these effects. The strategies developed in this paper are based on the KW theory and use the logic similar to Chen et al. (2014). However, the new strategies address more effectively several critical issues for non-recurrent severe BNs: (a) a very restrictive speed limit (lower than the speed in queue as in Chen et al. (2014)) should be avoided for better driver acceptance and compliance because congestion for this type of BN is likely more severe; (b) the upstream queue management should be more elaborate to provide smoother queue transition; and (c) it may not be straightforward to precisely estimate the stable maximum discharge rate, which can lead to underutilization of BN or re-emergent queue at the BN.

This study develops a theoretical framework for VSL control to improve the performance of severe BNs (or any BNs where low VSL is not desired, and gradual speed transition is preferred). This is an important contribution given that existing strategies were primarily designed to mitigate recurrent BNs. The VSL control strategies developed in this study are relatively simple, yet capable of addressing two critical issues for severe BNs: (i) reducing total delays significantly and (ii) providing smoother speed transition for better safety. The theoretical approach provides insights into traffic dynamics with VSL control and the impact of control parameters on the system performance (e.g., delay savings). Moreover, it provides a foundational framework to address more complex freeway networks and incorporate various implementation issues (e.g., detection and control technologies).

The remaining manuscript is organized as follows. Section 2 describes the basic VSL control strategy including the analysis of parameters on the system performance and sensitivity. Section 3 develops VSL strategies to address uncertainty in the stable maximum flow estimation. Concluding remarks are provided in Section 4.

2. Basic VSL control for severe bottlenecks

2.1. Baseline case

We study severe freeway BNs due to major road events that may partially block the roadways and reduce the throughputs, as shown by Fig. 1(a). We assume that the traffic evolution can be well approximated by the KW model with triangular fundamental diagrams (FD). The main logic of our VSL control, however, does not depend on the shape of FD, although some details of traffic evolution, such as shock wave speed, would change. The upper FD in Fig. 1(b) describes traffic states upstream of a BN with free-flow speed u, wave speed w and jam density k_j; and the lower FD describes traffic states at the BN location with lower free-flow speed u^{evt} and jam density k_j^{evt}. Note that we assume $u^{evt} < u$ due to rubbernecking and other effects induced by the road event.

We assume that traffic demand is constant in state A, and traffic breaks down to state H when the BN becomes active. After the road event is cleared, traffic recovers the full, normal capacity of q_{max} in state M. State e represents the stable maximum flow state that can be sustained at the BN without breakdown for an extended period; i.e., the BN capacity, $q_{BN} = q_e$. (The notation, $q_{_}$, represents flow in the traffic state denoted in subscript.) Note that $q_H < q_{BN}$ due to LC disruptions; and $q_{BN} - q_H$ represents the potential gain of system throughput. States G and E correspond respectively to the free-flow and congested states with the same flow as q_{BN} (i.e., $q_{BN} = q_E = q_G = q_e$) upstream of the BN.

The spatiotemporal traffic evolution without any control is illustrated in Fig. 1(c). After the road event, a heavy queue (in state H) propagates upstream, forming a shock wave, s_{AH}. (Hereafter, $s_{_}$ refers to a shock wave delineating two different traffic states denoted in subscript and represents the shock wave speed when used in equations.) When the road event is cleared at

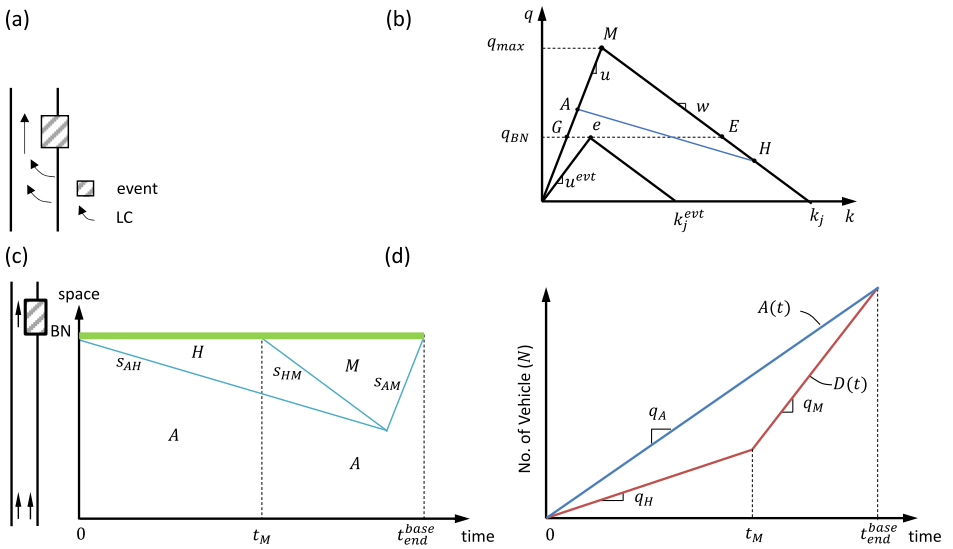


Fig. 1. Baseline case without VSL: (a) Illustration of lane-changing around road event; (b) Fundamental diagram; (c) Time–space diagram; (d) Queueing diagram.

time t_M, the normal capacity recovers and traffic evolves to state M, with the transition marked by s_{HM}. The queue ends when s_{AH} meets s_{HM}, at which state A is resumed; this demand recovery is marked by s_{AM}. The queue ending time, t_{end}^{base}, and the total delay resulting from the road event, W^{base}, can be easily derived from the queuing diagram in Fig. 1(d), in which $A(t)$ and $D(t)$ denote the virtual arrival and departure curves at the upstream starting point of the BN (the green bar on Fig. 1(c)), respectively:

$$t_{end}^{base} = \frac{(q_M - q_H)t_M}{q_M - q_A}, \tag{1}$$

$$W^{base} = \int (A(t) - D(t))dt = \frac{1}{2}(q_A - q_H)t_M t_{end}^{base} = \frac{(q_A - q_H)(q_M - q_H)t_M^2}{2(q_M - q_A)}. \tag{2}$$

(More detailed derivations of all equations in this paper are provided in Appendix A.) Notably, if the speed in queued state H is low, which is quite likely with a major incident or road construction, it would be undesirable to impose a restrictive speed limit ($<v_H$) to clear the queue around the BN as prescribed by Chen et al. (2014). Additionally, the speed drop along s_{AH} is likely abrupt, and the upstream queue management strategies in Chen et al. (2014) may not be sufficient to provide smooth enough transition at the queue's tail. Below we introduce a new VSL control strategy to address these problems.

2.2. Basic VSL control strategy

The control principle is simple: restrict the upstream demand (in free-flow) progressively. When done strategically, this will achieve three important objectives together: (i) to induce gradual transition at the queue's tail, (ii) to clear the queue around the BN, and (iii) to discharge traffic at stable maximum flow, q_{BN}, without breakdown. The control procedure consists of the following steps. All traffic states resulting from the basic VSL control are shown on the FD in Fig. 2(a), and the corresponding traffic evolution is shown in the time-space diagram in Fig. 2(b). All key variables will be derived in the parameter analysis section (Section 2.3).

Step 1: This main step is to control the upstream demand and clear the queue around the BN.

Set several intermediate speed limit values between u and v_E with an even increment. (The notation, $v_{\cdot}$, represents speed in the traffic state denoted in subscript.) The number of speed limit values will depend on the difference between u and v_E and the increment deemed acceptable to drivers. For an illustration purpose, we assume three intermediate values, denoted as V_1, V_2, and V_3.

Step 1-1: At the start of control (t_1), impose V_1 simultaneously over an extended segment immediately upstream of the queue; see Fig. 2(b). The length of the segment will be discussed later. This results in a zone with state $\tilde{a_1}$ (labeled as zone '1' in the figure), which has the same density as A, but the speed equals to V_1; see the FD in Fig. 2(a). Since the control is imposed simultaneously in space, the transition between A and $\tilde{a_1}$ forms a vertical shock, $s_{A\tilde{a_1}}$. At the downstream end of the control, state $\tilde{a}_1$ interacts with state H and forms $s_{\tilde{a}_1 H}$. Notice that with this control, the queue propagates more slowly; i.e., $|s_{\tilde{a}_1 H}| < |s_{AH}|$.

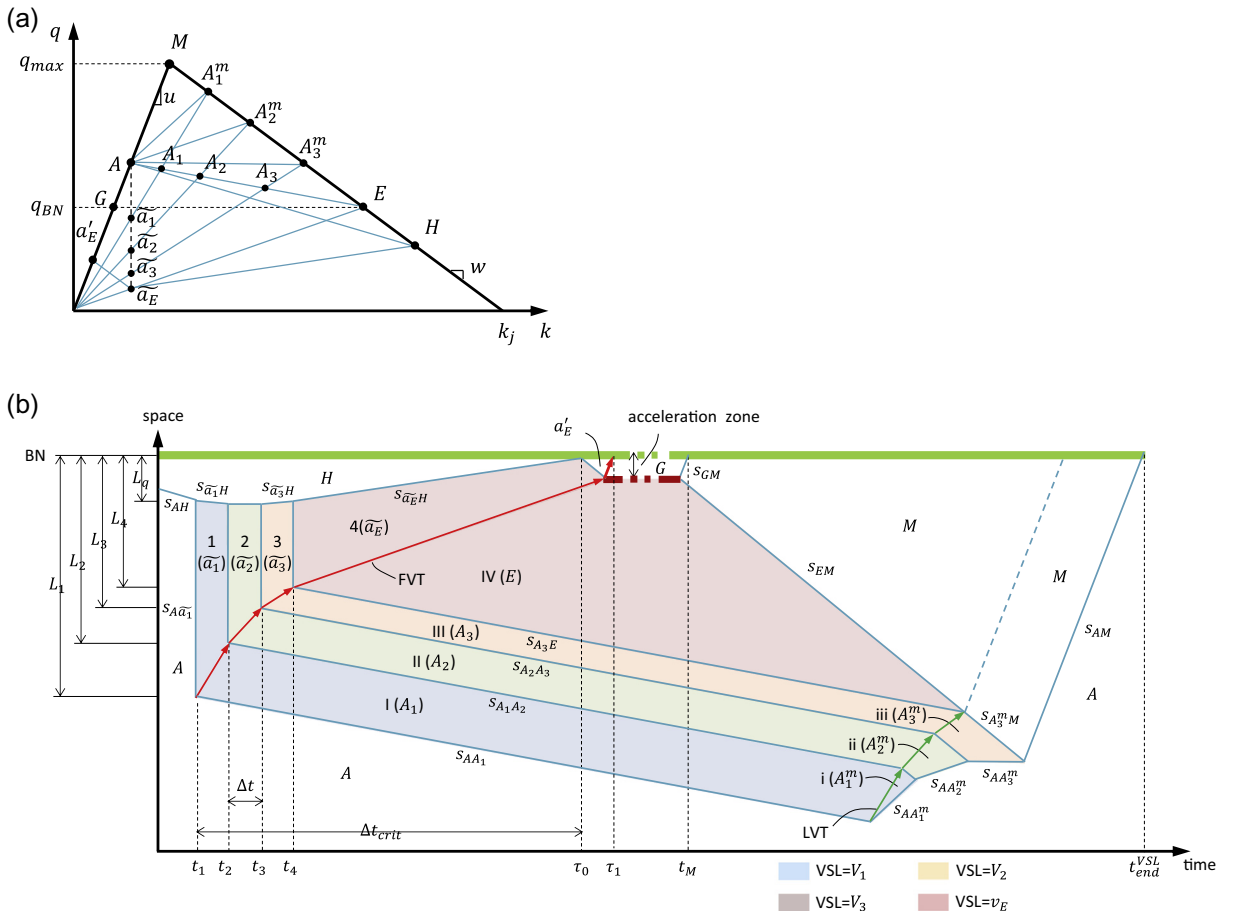


Fig. 2. Basic VSL control strategy: (a) Fundamental diagram illustrating traffic states that arise with VSL; (b) Time–space diagram illustrating shock waves generated by VSL; In (a), the fundamental diagram at the event site is omitted for better clarity of the figure; In (b), dotted line is used to indicate that the duration of state G was shortened to show the complete evolution.

Step 1-2: Switch the speed limit from V_1 to V_2 at time t_2 to create zone 2. Again, the new speed limit is actuated simultaneously in space immediately upstream of the queue. Similar to zone 1, zone 2 is in state $\widetilde{a_2}$ with the same density as A but with speed V_2 and forms $s_{\widetilde{a_2}H}$ where it meets state H. Notice that the queue propagates even more slowly; i.e., $\left|s_{\widetilde{a_2}H}\right| < \left|s_{\widetilde{a_1}H}\right| < |s_{AH}|$.

Step 1-3: Similar to Step 1-2, but change speed limit from V_2 to V_3 at time t_3 to create zone 3.

Step 1-4: Similar to Step 1-2, but change speed limit from V_3 to v_E at time t_4 to create zone 4 in state $\widetilde{a_E}$. Finally, the queue moves forward since $s_{\widetilde{a_E}H} > 0$ and is resolved[1] when $s_{\widetilde{a_E}H}$ arrives at the BN at τ_0.

Step 2: This step is to discharge traffic at the stable maximum flow, q_{BN}, without breakdown. The main idea is to have upstream traffic (in state A) gradually evolve to state E, and then let them fully accelerate to state G (or guide them to state e) before passing the BN.

Create an acceleration zone immediately upstream of the BN, where traffic can accelerate to u. The length of this zone, L, is set to be 0.35–1 km based on a typical vehicle acceleration capability as in Chen et al. (2014).

Step 2-1: When H is resolved at τ_0, de-activate v_E at the rate of the maximum backward moving wave speed, w, to guide the traffic in state $\widetilde{a_E}$ to accelerate to state a'_E until the shock reaches the entrance of the acceleration zone; see Fig. 2(b). Note that $q_{a'_E}$ may be smaller than q_{BN}, resulting in under-utilization of the BN capacity. Fortunately, this period is short $\left(=\frac{L}{u}-\frac{L}{w}\right)$ and negligible compared to the BN duration. For example, it lasts about 4 min for $u = 100$ km/h, $w = -18$ km/h, and $L = 0.75$ km. Nevertheless, one possible remedy is to increase the de-activation rate to achieve a higher discharge rate, which may be possible given the low density in $\widetilde{a_E}$ (relative to the FD). For simplicity when calculating the delay saving, we assume $q_{a'_E} = q_H$, which corresponds to a de-activation rate, w':

[1] For very low q_H, it is possible that $s_{\widetilde{a_E}H} < 0$. In this case, one can impose a stricter VSL ($<v_E$) to enable a forward shock. This would result in some underutilization of BN capacity later in the discharge process; however, this can be remedied by the strategy presented in Section 3.1.

$$w' = \frac{q_{a'_E} - q_{\widetilde{a_E}}}{k_{a'_E} - k_{\widetilde{a_E}}} = \frac{q_H - \frac{wq_E q_A}{(q_E + k_j w)u}}{\frac{q_H}{u} - \frac{q_A}{u}}. \quad (3)$$

The trajectory of the first vehicle that crosses the whole acceleration zone at u is denoted by the connected red arrows in Fig. 2(b), referred to as the first vehicle trajectory (FVT). The FVT serves as the state boundary and is used to back-calculate sequentially the lengths of the vertical control at v_E, V_3, V_2, and V_1, denoted as L_4, L_3, L_2, and L_1, respectively, as pictured in Fig. 2(b). Note that these lengths are measured from the bottleneck location rather than from the locations of shocks $s_{\widetilde{a_1}H} - s_{\widetilde{a_3}H}$ because (i) $V_1 - V_3$ and v_E are all greater than v_H, and therefore, would not impact traffic in state H, and (ii) using a fixed reference point is less cumbersome for implementation.

Step 2-2: Starting at the position of the FVT at t_1, impose V_1 at the rate of s_{AA_1} (=s_{AE}); see Fig. 2(a) for the FD and Fig. 2(b) for traffic evolution. At the same rate, impose V_2, V_3, and v_E when the FVT intersects t_2, t_3, and t_4, respectively, such that all shocks are parallel. Speed limit v_E is extended to the entrance of the acceleration zone.

This step results in four zones: zones I-IV, characterized by states $A_1 - A_3$ and E, respectively. Particularly, state A_1 in zone I has the same speed as state $\widetilde{a_1}$ but a higher density. Notice that in zones I through IV, vehicles gradually decrease their speed from u to v_E and then maintain v_E until reaching the entrance of acceleration zone. Thereafter, they resume u and traffic evolves to state G. Notice that state G eventually evolves to state e at the BN without any flow change, as prescribed by the lower FD; refer back to Fig. 1(b). (Recall that the lower free-flow speed at the BN is to capture rubbernecking and other effects induced by the road event.) In practice, this state change (from G to e) around the BN may instigate instability, especially given that some vehicles will have to merge to available lanes. In anticipation of this, traffic may alternatively be controlled at state e throughout the acceleration zone to attain more uniform and stable flow and speed. Notice that in this alternative case, the FVT in the acceleration zone (see Fig. 2(b)) will have speed v_e (=u^{evt}) rather than u. Regardless, the BN starts to discharge traffic at q_{BN} ($q_e = q_G = q_E$) after the arrival of the FVT.

Step 2-3: When the event is cleared at t_M, de-activate VSL to restore the full capacity of q_{max} (=q_M). To restore q_{max} as soon as possible, de-activation at the entrance of the acceleration zone can be timed so that the forward moving shock, s_{GM}, will reach the BN at t_M (as pictured in Fig. 2(b)). However, the gain is only about 1 minute (the trip time of s_{GM}), and thus, one may safely start the de-activation at t_M. Concurrently, VSL is de-activated at the rate of w, forming s_{EM}, which eventually collides with s_{A_3E} and terminates zone IV. Notably, the trajectory of the last vehicle that experiences state E is denoted by the three connected green arrows in Fig. 2(b), referred to as the Last Vehicle Trajectory (LVT). The LVT will serve as the boundary for VSL de-activation in the next step.

Step 2-4: De-activate V_1, V_2, and V_3 along the LVT to end the VSL control. As a result, three new zones, zones i–iii, form naturally (i.e., no VSL control is needed[2]) upstream of the LVT. States $A_1 - A_3$ evolve to congested states $A_1^m - A_3^m$, respectively, due to the bounded speeds, $V_1 - V_3$, downstream. When these states interact with the upstream demand, shocks $s_{AA_1^m} - s_{AA_3^m}$ form, which mark diminishing queue. As traffic emerges from state A_3^m, it evolves to the full capacity state, M, forming $s_{A_3^m M}$. When it meets $s_{AA_3^m}$, the queue completely vanishes. Note that the speed transitions around $s_{AA_1^m} - s_{AA_3^m}$ might be sharp, particularly around $s_{AA_3^m}$. Fortunately, the duration of this transition is usually short (about 1-3 minutes for typical parameter values). Alternatively, rather than de-activating VSL along the LVT, one can continue to de-activate VSL at the rate of w (= s_{EM}) after state E (zone IV) is terminated until zones III, II, and I are terminated successively. With this alternative solution, speed transitions would be smoother, albeit at the expense of bottleneck underutilization because states $A_3 - A_1$ would evolve to free-flow states with flows smaller than q_M.

2.3. Parameter analysis

Here we derive the critical control variables, the lengths of vertical control (from the bottleneck), $L_1 - L_4$. For this, we first derive Δt_{crit}, the duration of queue clearance at the BN after VSL; see the label in Fig. 2(b). For simplicity, we assume that the durations of zones 1–3 all equal to Δt. Then, Δt_{crit} is derived based on L_q, $s_{\widetilde{a_1}H}, s_{\widetilde{a_2}H}, s_{\widetilde{a_3}H}$, and $s_{\widetilde{a_E}H}$:

$$L_q = \left(s_{\widetilde{a_1}H} + s_{\widetilde{a_2}H} + s_{\widetilde{a_3}H}\right)\Delta t + s_{\widetilde{a_E}H}(\Delta t_{crit} - 3\Delta t).$$

Expressing the shocks in terms of the parameters given, and then solving for Δt_{crit}, we obtain:

$$\Delta t_{crit} = -\frac{3q_A w\Delta t(q_{BN}(u-w) + k_j wu) + 2L_q(q_{BN} + k_j w)(q_H u - q_A w + k_j wu)}{2w(-q_{BN}q_H u + q_A q_{BN} w - q_H k_j uw)}. \quad (4)$$

There are several factors that affect Δt_{crit}, namely Δt, the number of intermediate VSL values, and the most restrictive VSL, v_E. Clearly, the sooner we start v_E (i.e., Δt is smaller), the sooner we can clear the queue at the BN. Fig. 3 illustrates the impact, a linear positive relationship between Δt_{crit} and Δt, which increases faster with larger capacity drop, $1 - q_H/q_M$. Note that Δt should be sufficiently long so that drivers can adapt and smoothly change their speeds, which indicates the trade-off between the delay saving and speed transition. In the field implementation of SPECIALIST (Hegyi and Hoogendoorn,

[2] In practice, one may de-activate V_1, V_2, and V_3 along shocks $s_{AA_1^m} - s_{AA_3^m}$ to enhance the stability of traffic conditions within zones i, ii, and iii.

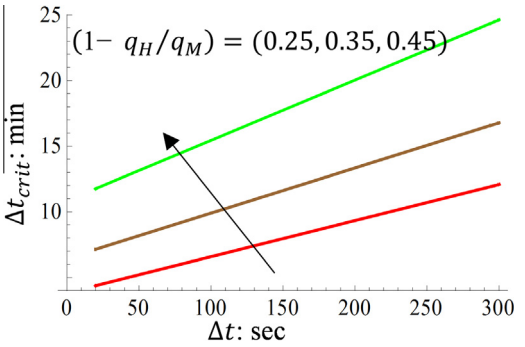


Fig. 3. Effect of capacity drop ($1 - q_H/q_M$) and zone 1–3 duration (Δt) on the duration of queue clearance at the bottleneck (Δt_{crit}).

2010), Δt = 15–20 s. Similarly, the number of intermediate VSL values between u and v_E can affect Δt_{crit}: given Δt, we can start v_E sooner with fewer intermediate VSL values. This again highlights the trade-off between the delay saving and speed transition. Finally, the effect of v_E on Δt_{crit} is intuitive: increasing v_E increases Δt_{crit} since VSL becomes less restrictive.

Based on Δt_{crit} and the speeds of the FVT, $L_1 - L_4$ can be derived as:

$$L_4 = (\Delta t_{crit} - 3\Delta t) * v_E - \frac{L}{w'}(v_E - w'), \tag{5-1}$$

$$L_3 = L_4 + V_3\Delta t = L_4 + V_3\Delta t, \tag{5-2}$$

$$L_2 = L_3 + V_2\Delta t = L_4 + (V_2 + V_3)\Delta t, \tag{5-3}$$

$$L_1 = L_2 + V_1\Delta t = L_4 + (V_1 + V_2 + V_3)\Delta t. \tag{5-4}$$

Turning attention to the delay saving with VSL, the new queuing diagram with the basic VSL control is shown in Fig. 4(a). Notice that the BN discharge rate increases to q_{BN} at τ_1, resulting in quicker overall queue dissipation at t_{end}^{VSL}. τ_1 can be derived based on u and w' from the time-space diagram in Fig. 2(b):

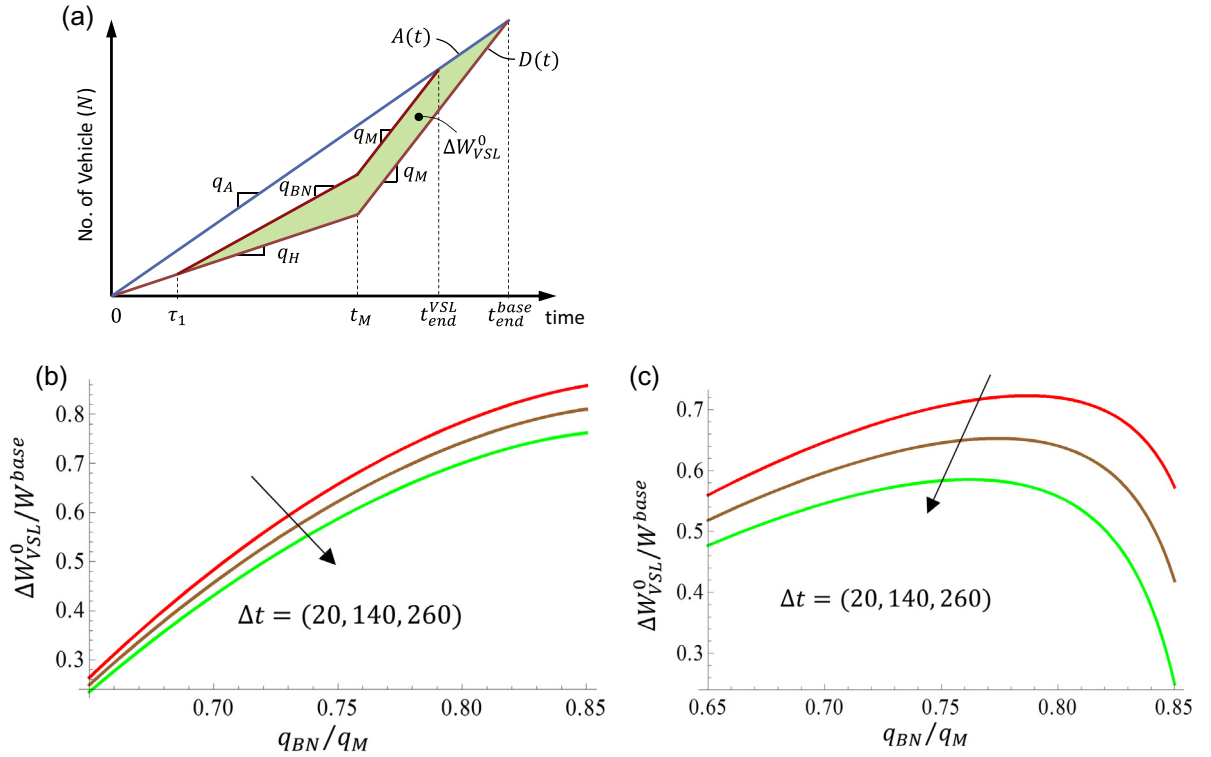


Fig. 4. Parameter analysis of basic VSL control strategy: (a) Queueing diagram illustrating delay saving; (b) Effect of fractional bottleneck capacity (q_{BN}/q_M) on the fractional delay saving $\left(\Delta W_{VSL}^0/W^{base}\right)$ for moderate non-recurrent congestion ($q_H = 0.6q_M$); (c) Effect of q_{BN} on $\Delta W_{VSL}^0/W^{base}$ for severe congestion ($q_H = 0.45q_M$); (For (b-c): $q_A = 0.8q_M$, L_q = 2 km, L = 0.75 km, T_M = 2 h).

$$\tau_1 = \tau_0 + L\left(\frac{1}{u} - \frac{1}{w'}\right) = -\frac{L_q}{s_{AH}} + \Delta t_{crit} + L\left(\frac{1}{u} - \frac{1}{w'}\right). \tag{6}$$

t_{end}^{VSL} can be derived from the queueing diagram in Fig. 4(a):

$$t_{end}^{VSL} = \frac{(q_{BN} - q_H)\tau_1 + (q_M - q_{BN})t_M}{(q_M - q_A)}. \tag{7}$$

Then, the delay saving as a result of the basic VSL control, ΔW_{VSL}^0, equals to the area of the shaded region in Fig. 4(a):

$$\Delta W_{VSL}^0 = 0.5 * (q_{BN} - q_H)(t_M - \tau_1)\left(\left(t_{end}^{VSL} - \tau_1\right) + \left(t_{end}^{base} - t_M\right)\right). \tag{8}$$

Fig. 4(b) and (c) illustrate the impact of q_{BN} on ΔW_{VSL}^0 for different levels of non-recurrent congestion. They are respectively expressed as the fractions relative to the road capacity and baseline total delay (i.e., q_{BN}/q_M vs. $\Delta W_{VSL}^0/W^{base}$) to better examine the relative impact. In moderate non-recurrent congestion ($q_H = 0.6q_M$; Fig. 4(b)),[3] $\Delta W_{VSL}^0/W^{base}$ increases with respect to q_{BN}/q_M, albeit at decreasing rates, representing diminishing marginal delay saving. In severe congestion (e.g., $q_H = 0.45q_M$), the fractional delay saving increases with q_{BN}/q_M up to a certain point and then decreases markedly; see Fig. 4(c). This trend of diminishing marginal delay saving is attributable to the counteraction between τ_1 and the discharge rate thereafter, as a result of setting the speed limit at v_E. Specifically, increasing v_E delays τ_1; however, as evident from Fig. 4(a), the BN would discharge traffic at a higher rate with a higher v_E (recall that $q_{BN} = q_E$) after τ_1, leading to faster overall queue clearance. The result in Fig. 4 indicates that the impact of delayed τ_1 on the delay saving becomes much greater than the impact of the queue dissipation rate as q_{BN}/q_M increases. Also notice that the impact of Δt is quite significant. For example, when Δt increases from 20 s to 140 s, the delay saving can drop by more than 10%; see Fig. 4(c).

3. VSL control with uncertainty in the stable maximum flow estimation

Uncertainty in estimation of the stable maximum flow, q_{BN}, is a valid concern due to the non-recurrent, wide-varying nature of the BNs that are considered in this study. Moreover, difficulty in predicting rubbernecking behavior adds to the challenge. If q_{BN} is underestimated, the BN would be underutilized. Conversely, an overestimated q_{BN} would trigger a queue at the BN, and the discharge rate would revert to q_H, resulting in no delay savings. In this section, we analyze the two cases, underestimated and overestimated q_{BN}, and develop VSL control strategies to remedy the loss accordingly.

3.1. Underestimated q_{BN}: Traffic evolution and VSL control

With an underestimated q_{BN}, some delay saving would be realized as long as $q_{BN} > q_H$. Thus, one may start with a conservative estimate of q_{BN} ($>q_H$) and gradually increase it toward the actual/optimal value, q_{BN^*}, to better utilize the BN. Specifically, after the initial queue is cleared, the BN discharge rate can be gradually increased: i.e., $q_{BN_0} \rightarrow q_{BN_1} \rightarrow q_{BN_2} \cdots \rightarrow q_{BN_n}$, where $q_{BN_0} < q_{BN_1} < q_{BN_2} \cdots < q_{BN_n} \leqslant q_{BN^*}$ and n is the number of updates. This can be implemented in Step 2–2 of the basic VSL control strategy, in which the initial queue is cleared and traffic is discharged at q_{BN}. Fig. 5 illustrates this strategy for $n = 2$; see Fig. 5(a) for the traffic states and Fig. 5(b) for the traffic evolution. Initially, VSL is imposed at v_E upstream of the acceleration zone to control the BN discharge rate at q_{BN_0} ($=q_E$). Soon after, VSL is changed to v_{E_1} and then v_{E_2} at times t_{E_1} and t_{E_2}, respectively, to increase the BN discharge rates to q_{BN_1} and q_{BN_2}. The latter VSL control (v_{E_2}) continues until the road event is cleared at time t_M.

These updates result in two additional zones upstream: zones IV-1 and IV-2 characterized by state E_1 and E_2, respectively. Therefore, traffic emerges from zone IV increases their speed to v_{E_1} in zone IV-1 and then to v_{E_2} in zone IV-2. After entering the acceleration zone, traffic in E_1 and E_2 evolves to G_1 ($= q_{BN_1}$) and G_2 ($= q_{BN_2}$), respectively. Upstream, states E_1 and E_2 interact with state A_3, resulting in shocks $s_{E_1E_2}$ and $s_{A_3E_2}$. When these shocks collide with s_{E_1M} and s_{E_2M}, zones IV-1 and IV-2 are terminated successively.

3.2. Underestimated q_{BN}: Parameter analysis

The queueing diagram with the "upward" adjustment is shown in Fig. 6(a). At τ_1 the BN discharge flow increases from q_H to q_{BN_0}, realizing some delay saving. More delay saving is achieved when the discharge flow increases to q_{BN_1} and q_{BN_2} later at τ_{BN_1} and τ_{BN_2}, respectively. In the general case with n updates, the queue clearance time, t_{end}^U, and total delay saving (compared to the base case), ΔW_{VSL}^U, are given by:

$$t_{end}^U = \frac{q_M t_M - q_H \tau_{BN_0} - \sum_0^n q_{BN_i}(\tau_{BN_{i+1}} - \tau_{BN_i})}{q_M - q_A}, \tag{9}$$

$$\Delta W_{VSL}^U = \frac{1}{2}\sum_0^n (N_i + N_{i+1})(\tau_{BN_{i+1}} - \tau_{BN_i}) + \frac{1}{2}N_{n+1}\left[\left(t_{end}^{base} - t_M\right) + \left(t_{end}^U - t_M\right)\right], \tag{10}$$

[3] We caution that this level of congestion is "moderate" for a typical non-recurrent BN, but still much heavier than a typical recurrent BN where $q_H \approx 0.9q_M$.

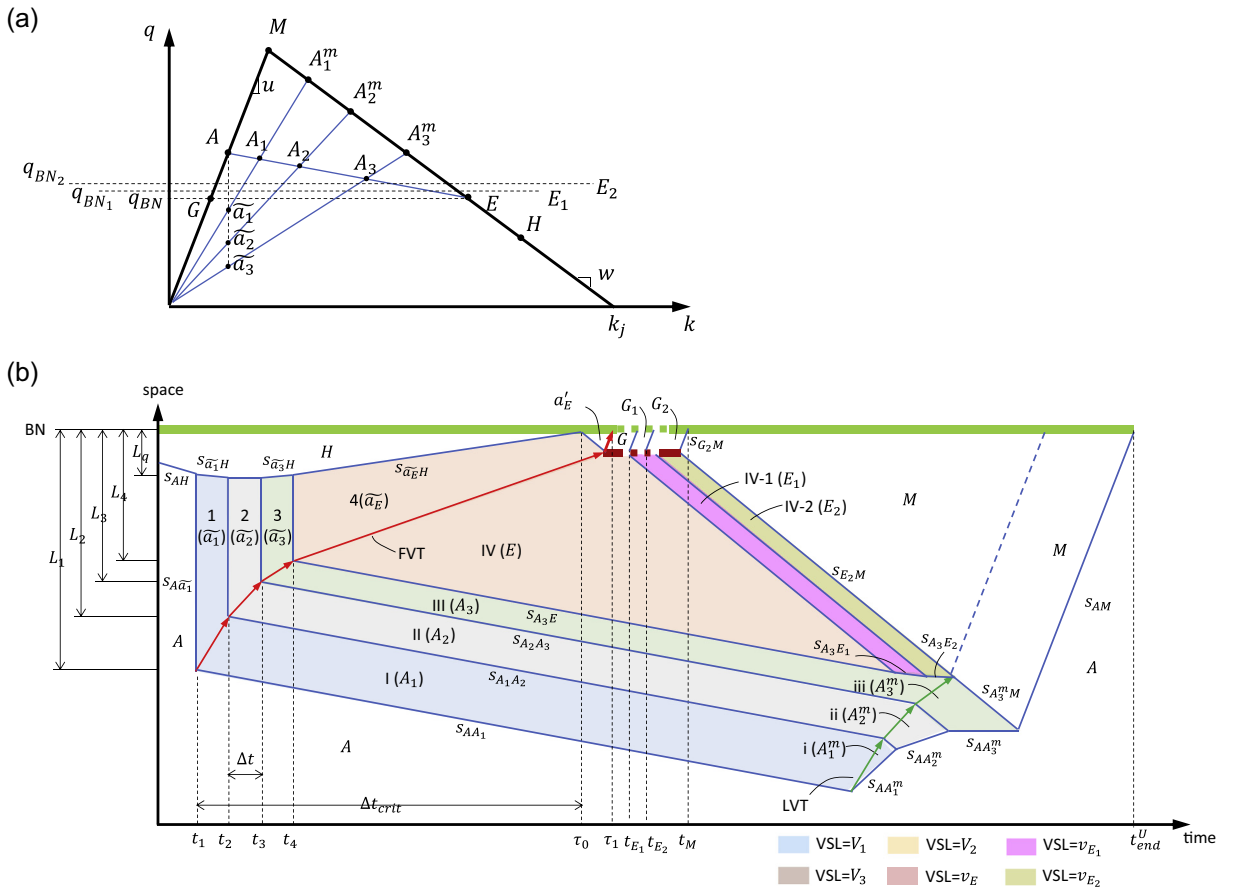


Fig. 5. VSL control to address underestimated q_{BN}: (a) Fundamental diagram; (b) Time–space diagram.

where τ_{BN_i} is when the BN discharge rate switches to q_{BN_i}; $\tau_{BN_{n+1}}$ equals to t_M; and N_i is the reduction in queue length at time τ_{BN_i} given by:

$$N_{i+1} = N_i + \sum_0^i (q_{BN_j} - q_H)(\tau_{BN_{j+1}} - \tau_{BN_j}), 0 \leqslant i \leqslant n,$$

$$N_0 = 0.$$

For the example illustrated in Section 3.1 ($n = 2$), $\tau_{BN_0} = \tau_1, \tau_{BN_1} = t_{E_1} + \frac{L}{u}$, and $\tau_{BN_2} = t_{E_2} + \frac{L}{u}$.

Now we show the impact of underestimation of q_{BN} on the loss of delay saving. The loss, ΔW_{VSL}^{ULoss}, is defined as $\Delta W_{VSL}^{*} - \Delta W_{VSL}^{U}$, where ΔW_{VSL}^{*} is the delay saving assuming q_{BN^*} throughout from the very beginning; see Fig. 6(b) for the queueing diagram. ΔW_{VSL}^{*} is given by:

$$\Delta W_{VSL}^{*} = 0.5 * (q_{BN^*} - q_H)(t_M - \tau_1^*)\left((t_{end}^* - \tau_1^*) + (t_{end}^{base} - t_M)\right), \tag{11}$$

where τ_1^* is the time when q_{BN^*} is resumed after the initial queue clearance at the BN, and t_{end}^* is the corresponding time of overall queue clearance. To be concise, we do not show the detailed equation for ΔW_{VSL}^{ULoss} here, but it can be obtained easily using (10) and (11).

Fig. 6(c) and (d) show the relationship between the fractional loss of delay saving ($\Delta W_{VSL}^{ULoss}/W^{base}$) and the level of underestimation $\left(1 - q_{BN_0}/q_{BN^*}\right)$ for moderate congestion ($q_H = 0.6q_M$) and severe congestion ($q_H = 0.45q_M$), respectively. To simplify the parameter analysis, we set $q_{BN^*} = 0.75q_M, q_{BN_1} = \left(q_{BN^*} + q_{BN_0}\right)/2$, and $q_{BN_2} = q_{BN^*}$. We also assume that the durations of q_{BN_0} and q_{BN_1} are equal, denoted by Δt_E ($= \tau_{BN_1} - \tau_{BN_0} = \tau_{BN_2} - \tau_{BN_1}$). As expected, the loss of delay savings generally increase with respect to the level of underestimation, and greater loss is observed with greater Δt_E since q_{BN^*} is recovered later. As observed in the basic control case, the counteraction between τ_1 and q_{BN} is noticeable particularly for severe congestion: the marginal loss of delay saving increases with respect to the level of underestimation due to the greater effect

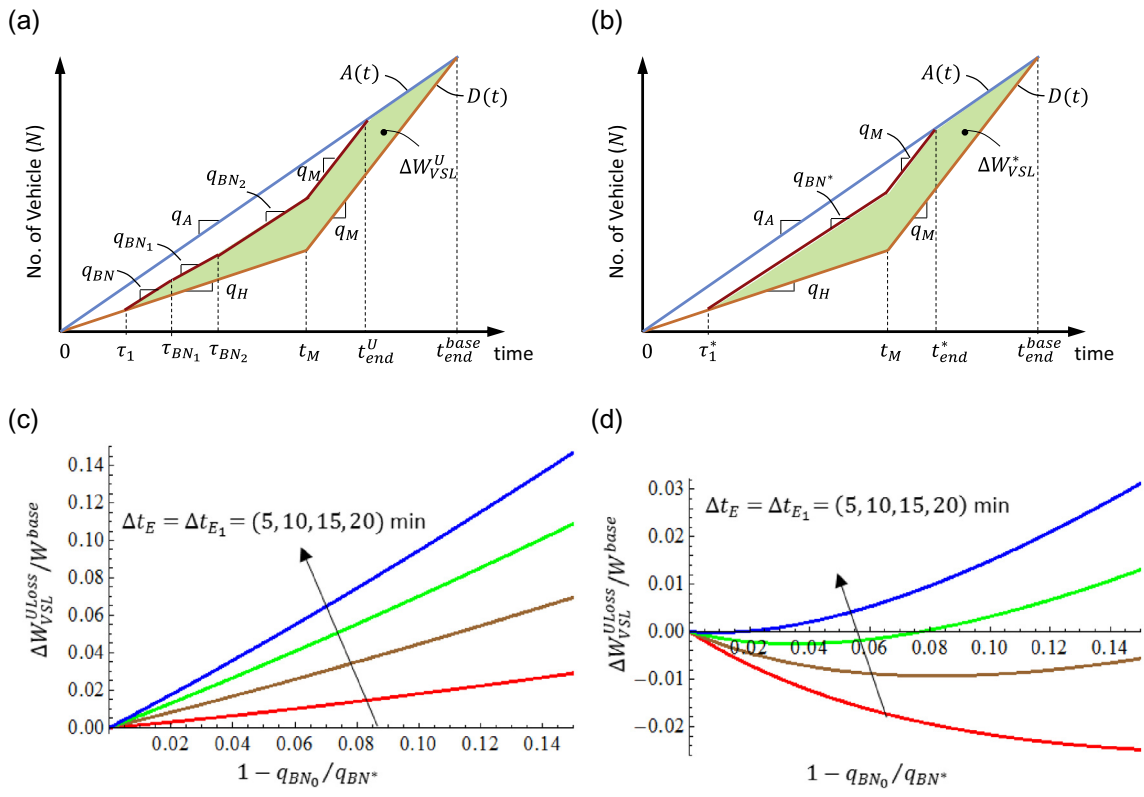


Fig. 6. Parameter analysis of VSL control with underestimated q_{BN}: (a) Queueing diagram illustrating delay saving with upward adjustment of q_{BN}; (b) Queueing diagram illustrating delay saving with optimal bottleneck discharge rate, q_{BN^*}; (c) Effect of underestimation level $(1 - q_{BN_0}/q_{BN^*})$ and Δt_E on the loss of fractional delay saving $\left(\Delta W^{ULoss}_{VSL}/W^{base}\right)$ for moderate non-recurrent congestion ($q_H = 0.6q_M$); (d) Effect of underestimation level $(1 - q_{BN_0}/q_{BN^*})$ and Δt_E on the loss of fractional delay saving $\left(\Delta W^{ULoss}_{VSL}/W^{base}\right)$ for severe congestion ($q_H = 0.45q_M$); (For (c-d): $q_A = 0.8q_M, L_q = 2$ km, $L = 0.75$ km, $T_M = 2$ h, $\Delta t = 20$ sec, $q_{BN^*} = 0.75q_M$, $q_{BN_2} = q_{BN^*}, q_{BN_1} = \left(q_{BN^*} + q_{BN_0}\right)/2$).

of the lower BN discharge rate than faster initial queue clearance (smaller τ_1). Moreover, for the severe congestion case (Fig. 6(d)), the loss is negative and exhibits decreasing trends for small Δt_E due to faster queue clearance (smaller τ_1) and recovery of q_{BN^*}. Nonetheless, the loss of fractional delay saving as a result of underestimating q_{BN} is overall not very significant (< 10% for moderate congestion and < 3% for severe congestion) if the VSL control is fairly responsive (Δt_E < 15 min).

3.3. Overestimated q_{BN}: Traffic evolution and VSL control

We now turn our attention to the case of an overestimated stable maximum discharge flow. We assume that the actual stable maximum flow (q_{BN^*}) is smaller than q_{BN} (and thus q_e, q_E and q_G). As a result, a queue forms again at the BN at τ_1(when the BN is supposed to start discharging at q_{BN})[4] and propagates as s_{GH}; see Fig. 7(a) for the FD and 7(b) for the traffic evolution. Hereon, traffic reverts to state H, and the BN discharges traffic at q_H rather than q_{BN}. If no action is taken, the new queue would continue to travel upstream, albeit more slowly than the initial queue ($s_{GH} < s_{AH}$), and eventually terminate transition zones I-IV. To remedy this problem, we impose new VSL control to clear the newly formed queue and then adjust the discharge flow to q_{BN^*} (or to a closer value).

This strategy has four steps with two objectives: to achieve q_{BN^*} as soon as possible and to smooth speed transition upstream. All traffic states resulting from this strategy are shown on the FD in Fig. 7(a), and the corresponding traffic evolution is shown in the time-space diagram in Fig. 7(b).

Step O1: When a new queue is confirmed (at t_V), impose new VSL control with speed limit v_H simultaneously over a segment immediately upstream of the acceleration zone to clear the queue; see Fig. 7(b). By flow conservation, a new state, $\widetilde{E}$, is created in zone V with the same density as state E. Here we assume that the length of zone IV (in state E) measured from the entrance of the acceleration zone, l_{IV}, is greater than the length of the required control, l_V, at t_V: i.e., $l_{IV} > l_V$. The contrary case of $l_{IV} < l_V$ is examined later. The transition between E and $\widetilde{E}$ forms a vertical shock, $s_{E\widetilde{E}}$. State $\widetilde{E}$ evolves to state I $(q_I = q_{\widetilde{E}})$ in the

[4] It is possible that a queue may form earlier if $q_{a'_E}$ is sufficiently high. However a VSL strategy to remedy this case would be similar to the strategy presented in this paper.

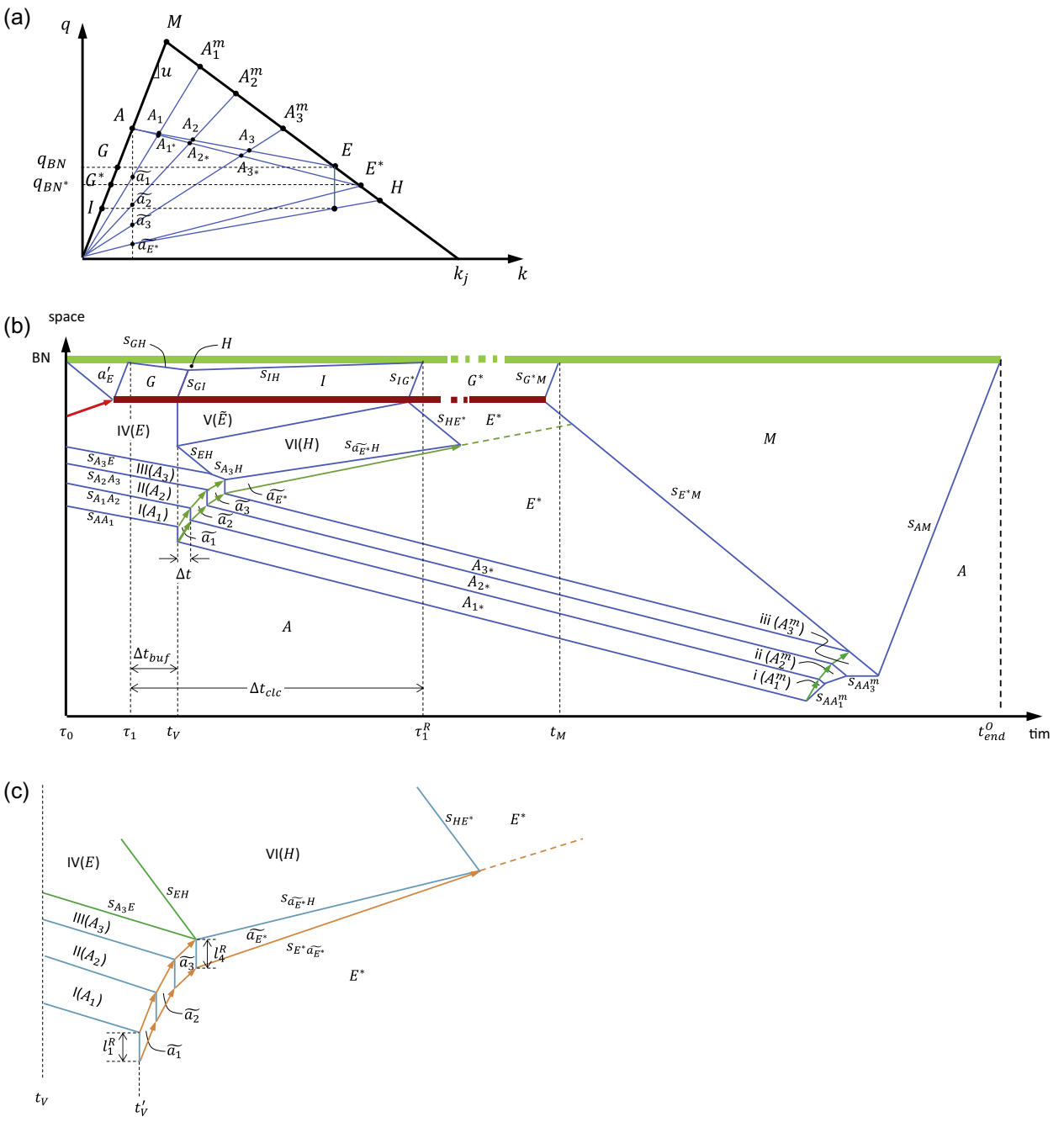


Fig. 7. VSL control to address overestimated q_{BN}: (a) Fundamental diagram; (b) Time–space diagram; (c) Enlarged time–space diagram for late termination of state *E*.

acceleration zone and then resolves the queue, forming a forward moving shock, s_{IH}. The arrival of s_{IH} at the BN marks the clearance of the new queue, which indicates that we can start discharging traffic at q_{BN^*}. l_V should be designed so that state *I* does not persist at the BN (as pictured). l_V will be derived in terms of parameters given, as part of the parameter analysis in Section 3.4.

Upstream of zone V, state *E* naturally evolves to state *H* (zone VI) and forms s_{EH}. Note that state *H* in zone VI is bounded by v_H due to state $\widetilde{E}$ downstream. Notably, s_{EH} is terminated as it travels upstream and collides with s_{A_3E}. Thereafter, state A_3 (zone III) interacts with *H*, forming s_{A_3H}.

Step O2: When the queue at the BN is cleared at τ_1^R, impose a less restrictive VSL, v_{E^*}, upstream of the acceleration zone to regulate the inflow to the BN at $q_{E^*}(=q_{BN^*})$. Upstream of the acceleration zone, traffic evolves from state *H* to E^*, forming s_{HE^*}, and then to state G^* in the acceleration zone. When the road event is cleared, this VSL control should be de-activated similar

to Step 2-3. Similar to the basic VSL strategy, one may time the actuation of v_{E^*} so that s_{IG^*} arrives at the BN at τ_1^R, or wait until the queue clearance to be safer.

For simplicity in implementation, the strategy can end with this step. However, without additional measures, the transition layers upstream (zones I-III) would vanish gradually, and the speed transition could become abrupt for a long enough duration. A more detailed discussion of traffic evolution under this simpler strategy is provided in Appendix A.1. Here we introduce additional steps to ensure smooth speed transition without affecting the delay savings.

Step O3: Impose VSL control on states A and $A_1 - A_3$ (transition zones I-III) to smooth speed transition. Specifically, starting at the same time with Step $O1$(at t_V), impose V_1 immediately upstream of state A_1 (zone I) over the same spatial extent[5] as A_1; see Fig. 7(a)-(b). Following V_1, impose V_2, V_3 and v_{E^*} sequentially with time increment of Δt. This creates low density and low flow states relative to the FD, $\widetilde{a_1} - \widetilde{a_3}$ and $\widetilde{a_{E^*}}$. The transition from $\widetilde{a_{E^*}}$ to H forms a forward shock, $s_{\widetilde{a_{E^*}}H}$, which will contain and eventually terminate state H when colliding with s_{HE^*}.

Note that state E may be terminated by state H early if s_{A_3E} and s_{EH} collide before v_{E^*} is supposed to go in effect, as depicted in Fig. 7(b). In this case, speed limit v_{E^*} is turned on sooner than Δt after V_3, more precisely, when $s_{A_3\widetilde{a_3}}$ collides with s_{A_3H}. By contrast, if state E still remains, the upstream control is delayed until t_V' so that the three shocks, s_{EH}, s_{A_3E}, and $s_{A_3\widetilde{a_3}}$, converge when v_{E^*} is imposed; see Fig. 7(c). In either case, the spatial extents of $V_1 - V_3$ and v_{E^*} at the start, $l_1^R - l_4^R$, respectively, should be determined such that state $\widetilde{a_{E^*}}$ is terminated simultaneously with shock s_{HE^*}; the derivations are provided in the following section. Note that $l_1^R = l_2^R = l_3^R = l_4^R$ since the trajectories of the first and last vehicles when passing states $\widetilde{a_1} - \widetilde{a_3}$ are parallel.

Step O4: Upstream of $\widetilde{a_1} - \widetilde{a_3}$ and $\widetilde{a_{E^*}}$, impose another set of $V_1 - V_3$ and v_{E^*} at the rate of s_{AE^*} to transition traffic gradually from A to E^* ($A \rightarrow A_{1*} \rightarrow A_{2*} \rightarrow A_{3*} \rightarrow E^*$); refer back to Fig. 7(b). Finally, de-activate the control when the road event is cleared, similar to Step 2-4.

Several extensions of the VSL control strategy are worth mentioning. Recall that the strategy described above assumes $l_{IV} > l_V$. However, the contrary is possible if the event-induced congestion is not severe (e.g., $q_H > 0.75q_M$). There are two options for this case. One is to set the speed limit below v_H such that $l_{IV} = l_V$. The required speed limit is derived in Appendix A.2. Traffic evolution for this option is qualitatively the same as Fig. 7(b), although shock wave speeds and the queue clearance time (τ_1^R) would be different (shown in Appendix A.2). If setting the speed limit below v_H is undesirable, another option is to simply extend the control (VSL=v_H) to zone III and on. Traffic evolution under this second option is different, and a more detailed description is provided in Appendix A.3.

3.4. Overestimated q_{BN}: Parameter analysis

This section derives the parameters used in the VSL control strategy in Section 3.3 and analyzes the effects of key parameters on the delay saving.

3.4.1. Parameters of VSL control

To determine the length of vertical control, l_V, we first derive Δt_{clc}, the duration of the new queue (labeled in Fig. 7(b)). Notice that Δt_{clc} depends on how responsive the system is. We assume that the new queue is detected after some buffer time, Δt_{buf}, or when the queue reaches the entrance of the acceleration zone, whichever happens first. Then, we find that Δt_{clc} is given by:

$$\Delta t_{clc} = \mathrm{Min}\left(-\frac{k_j w(L + u\Delta t_{buf})}{q_H u}, -\frac{L(q_H(u-w) + uk_j w)k_j}{(q_{BN} - q_H)q_H u}\right), \tag{12}$$

in which the former expression corresponds to early detection (i.e., after Δt_{buf}) and the latter for detection at the entrance. Notice that in the former case, a linear positive relationship exists between Δt_{clc} and Δt_{buf}, as shown in see Fig. 8. This is expected because the earlier we detect the queue, the sooner we can actuate the queue clearance process. In the latter case, Δt_{clc} depends on the initial estimation of the stable maximum flow. The higher the q_{BN} is, the smaller the Δt_{clc} is. This is because a higher q_{BN} results in faster s_{GH} and s_{IH}, leading to earlier queue detection at the entrance and faster queue dissipation. Notice that, as expected, Δt_{clc} does not depend on Δt_{buf} in the latter case; and this is reflected in the horizontal portions of Fig. 8.

Next, we determine l_V such that state $\widetilde{E}$ in zone V persists just long enough to clear the new queue, as pictured in Fig. 7(b). This duration of state $\widetilde{E}$, $\Delta t_{\widetilde{E}}$, is given by:

$$\Delta t_{\widetilde{E}} = \Delta t_{clc} - \Delta t_{buf} - \frac{L}{u} = \mathrm{Min}\left(-\frac{(L + u\Delta t_{buf})(q_H + k_j w)}{q_H u}, \frac{L(q_H - k_j w)(q_H(u-w) + uk_j w)}{(q_{BN} - q_H)q_H uw}\right), \tag{13-1}$$

Then,

$$l_V = \Delta t_{\widetilde{E}} v_H = \mathrm{Min}\left(-\frac{(L + u\Delta t_{buf})w}{u}, \frac{L(q_H - k_j w)(q_H(u-w) + uk_j w)}{(q_{BN} - q_H)u(q_H + k_j w)}\right). \tag{13-2}$$

[5] This is because the vehicle trajectory passing these three zones is parallel to those passing zones 1–3, such as the FVT.

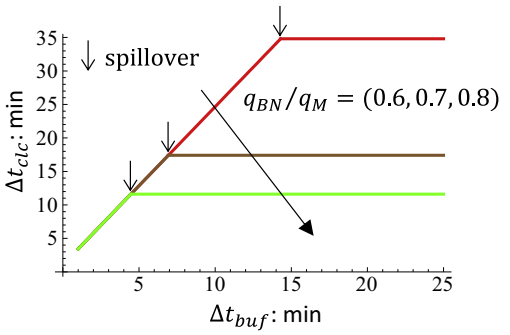


Fig. 8. Effect of buffer time of new queue detection (Δt_{buf}) on the duration of new queue clearance at the bottleneck ($q_H = 0.5q_M$, $L = 0.75$ km) (the arrow indicates the time when queue has spilled over the entrance of acceleration zone.).

Also from the geometry of Fig. 7(b), we can derive l_{IV}:

$$l_{IV} = L_4 - s_{A_3E}\left(\Delta t_{crit} - 3\Delta t - \frac{L}{w'} + \frac{L}{u} + \Delta t_{dt}\right) - L, \tag{13-3}$$

in which Δt_{dt} is the actual time when the queue is detected, given by:

$$\Delta t_{dt} = \text{Min}\left(\Delta t_{buf}, -\frac{L}{s_{GH}}\right).$$

As revealed by Eq. (13-1), in the case of early detection, $\Delta t_{\widetilde{E}}$ and l_V increase with Δt_{buf}. This is because shock s_{IH} has to travel longer. For the late detection (at the entrance to the acceleration zone), $\Delta t_{\widetilde{E}}$ and l_V increase as q_{BN} decreases because s_{IH} travels more slowly.

Recall that state E can be terminated by state H early (i.e., case of Fig. 7(b)) or late (i.e., case of Fig. 7(c)). In the latter case, the starting time of upstream control, t'_V, is given by:

$$t'_V = t_V + \frac{l_{IV} - l_V}{(s_{A_3E} - s_{EH})} - 3\Delta t, \tag{14}$$

in which the term $\frac{l_{IV}-l_V}{(s_{A_3E}-s_{EH})}$ is the additional time traveled by shocks s_{EH} and s_{A_3E} after t_V. The lengths of vertical control to provide smoother transition, l_4^R (and thus $l_1^R - l_3^R$), can be back-calculated based on the condition that $\widetilde{a_{E^*}}$ is terminated simultaneously with shock s_{HE^*}:

$$l_4^R = \left(v_{E^*} - s_{\widetilde{a_{E^*}H}}\right)\frac{\Delta t_{\widetilde{E}}(v_H - w)}{s_{\widetilde{a_{E^*}H}} - w}. \tag{15}$$

In the case that state E is terminated by state H early, l_4^R will be slightly smaller than (15) but the difference is negligible because the total duration of the three states $\widetilde{a_1} - \widetilde{a_3}$ is only $3\Delta t$, which is usually less than 1 minute. Thus (15) can be used without causing a significant loss of discharge flow.

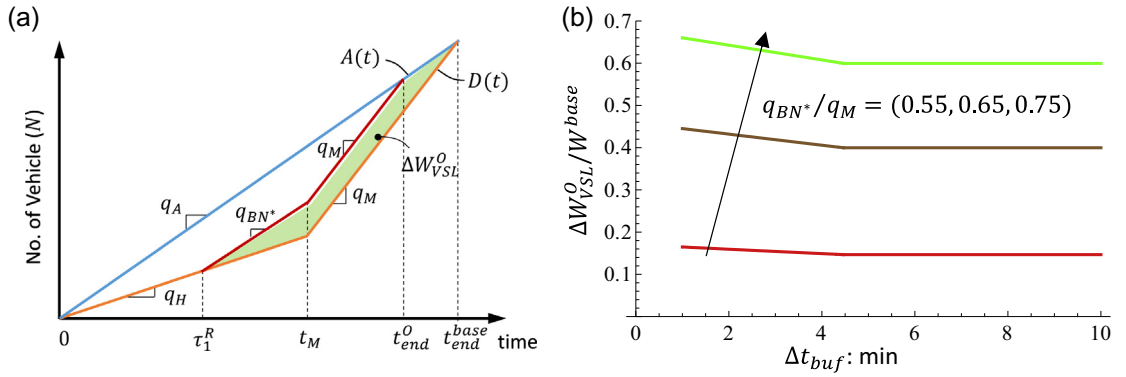


Fig. 9. Parameter analysis of VSL control to address overestimated q_{BN}: (a) Queueing diagram illustrating delay saving; (b) Effect of Δt_{buf} on the fractional delay saving $\left(\Delta W^O_{VSL}/W^{base}\right)$; ($q_A = 0.8q_M$, $q_{BN} = 0.8q_M$, $q_H = 0.5q_M$, $L_q = 2$ km, $L = 0.75$ km, $T_M = 2$ h).

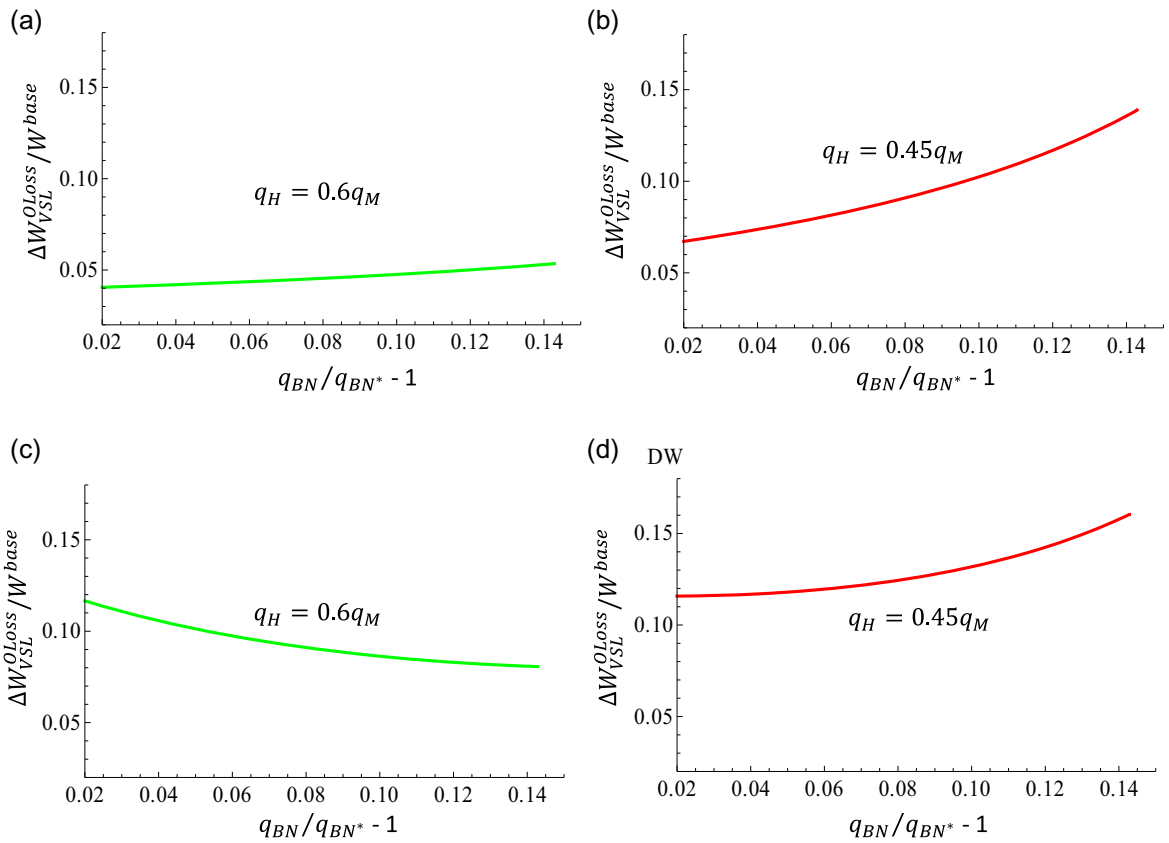


Fig. 10. The level of q_{BN} overestimation ($q_{BN}/q_{BN^*} - 1$) vs. loss of fractional delay savings $\left(\Delta W_{VSL}^{OLoss}/W^{base}\right)$: (a) Moderate non-recurrent congestion ($q_H = 0.6q_M$) for early detection of re-emergent queue ($\Delta\ t_{buf} = 3$ min); (b) Severe congestion ($q_H = 0.45q_M$) for early detection of re-emergent queue ($\Delta t_{buf} = 3$ min); (c) Moderate non-recurrent congestion ($q_H = 0.6q_M$) for late detection of re-emergent queue ($\Delta t_{buf} = 15$ min); (d) Severe congestion ($q_H = 0.45q_M$) for late detection of re-emergent queue ($\Delta\ t_{buf} = 15$ min); (For (a–d): $q_A = 0.8q_M, q_{BN^*} = 0.7q_M, \Delta t = 20$ sec, $L_q = 2$ km, $L = 0.75$ km, $T_M = 2$ h).

Note that in the strategy above, we use speed limit v_H (when assuming $l_{IV} > l_V$) to clear the newly formed queue at the BN. Obviously, the queue would dissipate faster with a lower speed limit. However, what is acceptable to drivers should determine the lower bound. The upper bound for the speed limit is more definite. The maximum value, v_{max}, is the speed that yields $q_I = q_H$:

$$v_{max} = \frac{q_H w}{q_{BN} + k_j w}. \tag{16}$$

A speed limit higher than v_{max} would result in $s_{IH} < 0$ and it would not be possible to clear the newly formed queue.

3.4.2. Analysis of delay saving

We now turn our attention to the queueing analysis. As expected, the delay saving would decrease if a queue re-emerges at the BN; see Fig. 9(a) for the queuing diagram. The delay saving (compared to the base case) is ΔW_{VSL}^{O} and can be derived based on the queueing diagram:

$$\Delta W_{VSL}^{O} = 0.5(q_{BN^*} - q_H)(t_M - \tau_1^R)\left((t_{end}^{O} - \tau_1^R) + (t_{end}^{base} - t_M)\right). \tag{17}$$

τ_1^R represents the time when q_{BN^*} is resumed and equals to $\tau_1 + \Delta t_{clc}$.

The relationship between the fractional delay saving $\left(\Delta W_{VSL}^{O}/W^{base}\right)$ and Δt_{buf} is illustrated in Fig. 9(b). With early detection, it decreases slightly as the buffer time increases, which is expected because Δt_{clc} is larger. Also notice that the delay saving depends on q_{BN^*}: obviously, a larger q_{BN^*} leads to a higher delay saving due to the increased discharge flow.

We also derive the loss of delay saving due to the re-emergent queue, ΔW_{VSL}^{OLoss}, defined as the difference of delay saving between ΔW_{VSL}^{O} and ΔW_{VSL}^{*}, where ΔW_{VSL}^{*} is the delay saving if the BN discharge rate is set to be q_{BN^*} from the very beginning[6] defined in Section 3.2 and shown by Fig. 7(a). Then using (11) and (17) ΔW_{VSL}^{OLoss} is derived as:

$$\begin{aligned}\Delta W_{VSL}^{OLoss} &= \Delta W_{VSL}^{*} - \Delta W_{VSL}^{O} \\ &= 0.5 * (q_{BN^*} - q_H)\left[(t_M - \tau_1^*)\left((t_{end}^* - \tau_1^*) + (t_{end}^{base} - t_M)\right) - (t_M - \tau_1^R)\left((t_{end}^O - \tau_1^R) + (t_{end}^{base} - t_M)\right)\right]. \end{aligned} \quad (18)$$

Fig. 10(a) and (b) illustrate the relationship between the fractional loss of delay savings $\left(\Delta W_{VSL}^{OLoss}/W^{base}\right)$ and the level of overestimation $(q_{BN}/q_{BN^*} - 1)$ with early detection of the re-emergent queue in moderate ($q_H = 0.6q_M$) and severe congestion ($q_H = 0.45q_M$), respectively. One can see that the fractional loss increases at an increasing rate with respect to the overestimation level. The increasing rate is particularly noticeable for severe congestion. This is mainly driven by the fact that a higher q_{BN} leads to a larger Δt_{crit} (the duration to clear the initial queue) because the speed limit at v_E becomes less restrictive, while Δt_{clc} (the duration to clear the re-emergent queue) is independent of q_{BN} for early detection (see the first term in Eq. (12)). Fig. 10(c) and (d) illustrate the same relationship for late detection (at the entrance of the acceleration zone) of the re-emergent queue. Notice that in comparing early vs. late detection, the trends for severe congestion are similar (Fig. 10(b) vs. Fig. 10(d)); however, the trend for moderate congestion is decreasing for late detection (Fig. 10(c)), contrary to the early detection case (Fig. 10(a)). This is because in the late detection case, $\Delta\, t_{clc}$ inversely related to q_{BN} (see the second term in Eq. (12)), thus countering the effect of q_{BN} on $\Delta\, t_{crit}$. Evidently, this counter effect becomes more dominant in moderate congestion. It is worth noting that the fractional loss is generally significant in severe congestion (mostly >10%); while it is less significant in moderate congestion. This indicates that one may error on the side of underestimation when the congestion is severe.

4. Conclusions and discussions

In this paper, variable speed limit (VSL) strategies were developed based on the Kinematic Wave theory to increase discharge rates at severe freeway bottlenecks induced by major road events such as incidents and work zones and provide smoother speed transition upstream. Our main logic is to impose VSL control gradually on upstream demand to dissipate the queue around the bottleneck while inducing smoother speed transition at the queue's tail. After the queue clearance, VSL control continues to regulate inflow to the bottleneck so that the bottleneck can discharge traffic at the stable maximum rate without breakdown. This is accomplished without imposing overly restrictive speed limits – an important feature considering heavily congested situations. Our findings from the parameter analysis suggest that significant delay savings can be realized with our strategy.

We further developed two sequel VSL strategies to remedy (i) underutilization of bottleneck capacity due to an underestimated stable maximum flow and (ii) a re-emergent queue at the bottleneck due to an overestimated stable maximum flow. For the former, a simple upward correction strategy (i.e., gradual increase of estimated stable maximum flow) was introduced. The analysis of parameters suggested that the penalty in delay saving by underestimation was not significant (mostly < 10%) for reasonable parameter values and that the proposed strategy was effective in reducing the penalty. For the case of overestimation, the proposed strategy was designed to clear the re-emergent queue at the bottleneck and discharge traffic at the adjusted (lower) stable maximum rate while controlling the upstream queue for smoother transition. Not surprisingly, we found that delay savings, though still substantial, decreased (i.e., positive loss of delay saving) when the stable maximum flow was over-estimated. However, the proposed VSL strategy was effective in minimizing the loss in delay saving.

Building on the theoretical framework presented in this paper, future refinement includes developing discrete schemes to accommodate more complex scenarios, such as time varying demand and realistic freeway networks (with on and off-ramps), and incorporating emerging technologies such as the Connected Vehicle technology to VSL control. Furthermore, while the strategies presented in this paper show potential for effectively addressing severe non-recurrent bottlenecks, they should be further refined and evaluated based on careful field experiments and observations to address various implementation issues. Notably, traffic states under VSL likely depend on how drivers react to VSL control (e.g., compliance speed and compliance rate) and presence speed enforcement. A careful field experiment would be necessary to (i) systematically characterize traffic states (which may be in non-equilibrium) under our VSL strategies, (ii) tune certain control parameters accordingly, (iii) modify the strategies to error on the side of caution, and/or (iv) develop a control scheme with feedback control enabled by real-time detection of traffic states. Finally, a field testing would enable evaluating system robustness with respect to driver compliance and traffic detection/measurement errors, delays and resolution.

Acknowledgments

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[6] Note that in this case, there would not be a re-emergent queue.

Appendix A

Here we show the derivation of the equations.

1. Eq. (1): derived from the geometric relationship shown on Fig. 1(d).
2. Eq. (2): again derived from the geometric relationship shown on Fig. 1(d).
3. Eq. (3): we have $w' = \frac{q_{a'_E} - q_{\tilde{a}_E}}{k_{a'_E} - k_{\tilde{a}_E}}$ from the FD in Fig. 2(a). Then, plug in the following information:

$$k_{\tilde{a}_E} = k_A,$$
$$q_{\tilde{a}_E} = k_A v_E,$$
$$k_A = \frac{q_A}{u},\ q_{a'_E} = q_H,\ k_{a'_E} = \frac{q_H}{u},$$
$$v_E = \frac{wq_E}{(q_{E+k_j w})},$$
$$q_E = q_{\mathrm{BN}},$$

We have:

$$w' = \frac{q_{a'_E} - q_{\tilde{a}_E}}{k_{a'_E} - k_{\tilde{a}_E}} = \frac{q_H - \dfrac{wq_{\mathrm{BN}}q_A}{u(q_{\mathrm{BN}} + k_j w)}}{\dfrac{q_H}{u} - \dfrac{q_A}{u}}.$$

4. Eq. (4): we rearrange the following equation:

$$L_q = \left(s_{\tilde{a}_1 H} + s_{\tilde{a}_2 H} + s_{\tilde{a}_3 H}\right)\Delta t + s_{\tilde{a}_E H}(\Delta t_{crit} - 3\Delta t).$$

Then we have:

$$\Delta t_{crit} = \frac{L_q - \left(s_{\tilde{a}_1 H} + s_{\tilde{a}_2 H} + s_{\tilde{a}_3 H}\right)\Delta t + 3s_{\tilde{a}_E H}\Delta t}{s_{\tilde{a}_E H}}.$$

Note that the expression for the shocks are given by:

$$s_{\tilde{a}_1 H} = \frac{q_H - q_{\tilde{a}_1}}{k_H - k_{\tilde{a}_1}},$$
$$s_{\tilde{a}_2 H} = \frac{q_H - q_{\tilde{a}_2}}{k_H - k_{\tilde{a}_2}},$$
$$s_{\tilde{a}_3 H} = \frac{q_H - q_{\tilde{a}_3}}{k_H - k_{\tilde{a}_3}},$$

and $s_{\tilde{a}_E H} = \frac{q_H - q_{\tilde{a}_E}}{k_H - k_{\tilde{a}_E}}$.

We plug in the following equations to $s_{\tilde{a}_1 H}$:

$$k_H = \frac{q_H w}{q_H + k_j w},$$
$$k_{\tilde{a}_1} = k_A = \frac{q_A}{u},$$
$$q_{\tilde{a}_1} = k_{\tilde{a}_1} V_1,$$
$$V_1 = u - \frac{u - v_E}{4}.$$
$$v_E = \frac{wq_{BN}}{q_{BN} + k_j w}.$$

We have:

$$s_{\tilde{a}_1 H} = -\frac{uw\left(-4q_H + q_A\left(3 + \dfrac{wq_{BN}q_A}{u(q_{BN} + k_j w)}\right)\right)}{4(q_H u - q_A w + k_j w u)}.$$

Similarly, $s_{\tilde{a_2}H}$ and $s_{\tilde{a_3}H}$ can be obtained as well with the following equations plugged in:

$$k_{\tilde{a_2}} = k_{\tilde{a_3}} = k_{\tilde{a_E}} = k_A = \frac{q_A}{u},$$
$$q_{\tilde{a_2}} = k_{\tilde{a_2}} V_2,$$
$$q_{\tilde{a_3}} = k_{\tilde{a_3}} V_3,$$
$$q_{\tilde{a_E}} = k_{\tilde{a_E}} v_E,$$
$$V_2 = u - 2\frac{u - v_E}{4}.$$
$$V_3 = u - 3\frac{u - v_E}{4}.$$

With these shocks, we obtain Eq. (4).

5. Eq. (5): derived from the geometric relationship shown on Fig. 2(b).
6. Eq. (6): derived from the geometric relationship shown on Fig. 2(b).
7. Eq. (7): derived from the geometric relationship shown on Fig. 4.
8. Eq. (8): derived from the geometric relationship shown on Fig. 4.
9. Eq. (9): derived from the geometric relationship shown on Fig. 6(a).
10. Eq. (10): derived from the geometric relationship shown on Fig. 6(a).
11. Eq. (11): derived from the geometric relationship shown on Fig. 6(a).
12. Eq. (12): derived from the geometric relationship shown on Fig. 7(b).

Firstly, let L_q^R denote the distance that shock s_{IH} has travelled. According to the geometric relationship, we have:

$$\Delta t_{clc} = -\frac{L_q^R}{s_{GH}} + \frac{L_q^R}{s_{IH}}.$$

In order to obtain L_q^R, we found that for the case of early detection, we have:

$$-\frac{L_q^R}{s_{GH}} - \frac{L_q^R}{u} = \Delta t_{buf}.$$

Solving the equation, we obtain:

$$L_q^R = -\frac{\Delta t_{buf}}{\frac{1}{s_{GH}} + \frac{1}{u}}.$$

In the case of late detection, L_q^R in either case: $L_q^R = L$.
Thus, with both cases combined, we have:

$$L_q^R = Min\left(-\frac{\Delta t_{buf}}{\frac{1}{s_{GH}} + \frac{1}{u}}, L\right).$$

Secondly, in addition to L_q^R, we plug in the expression for s_{GH} and s_{IH} as follows:

$$s_{GH} = \frac{q_G - q_H}{k_G - k_H}, \text{and } s_{IH} = \frac{q_I - q_H}{k_I - k_H},$$

in which

$$q_G = q_E = q_{BN}$$
$$k_G = \frac{q_G}{u}$$
$$q_I = k_E v_H$$
$$k_I = \frac{q_I}{u}.$$

Therefore, we obtain Eq. (9):

$$\Delta t_{clc} = \text{Min}\left(-\frac{k_j w(L + u\Delta t_{buf})}{q_H u}, -\frac{L(q_H(u - w) + uk_j w)k_j}{(q_{BN} - q_H)q_H u}\right).$$

13. Eq. (13): from the geometry on Fig. 7(b), we have:

$$\Delta t_{\tilde{E}} = \Delta t_{clc} - \Delta t_{buf} - \frac{L}{u},$$
$$l_V = \Delta t_{\tilde{E}} v_H.$$

Then we plug in Eq. (12) to obtain Eqs. (13-1) and (13-2).
For (13-3), from the geometry of Fig. 7(b), we have:

$$l_{IV} = L_4 - s_{A_3E}\left(\Delta t_{clc} - 3\Delta t - \frac{L}{w} + \frac{L}{u} + \Delta t_{dt}\right) - L.$$

The complete form of l_{IV} can be obtained by plug in L_4 and s_{A_3E}:

$$s_{A_3E} = s_{AE} = \frac{q_A - q_E}{k_A - k_E} = \frac{q_A - q_{BN}}{\frac{q_A}{u} - \frac{q_{BN}w}{q_{BN}+k_j w}}.$$

However, the complete form is excessively long without providing additional insights and thus, is not provided.

14. Eq. (14): derived from the geometric relationship shown on Fig. 7(b) or (c).
According to the geometry in Fig. 7(c), the additional time that shocks s_{EH} and s_{A_3E} travel to meet after t_V is:

$$\frac{l_{IV} - l_V}{(s_{A_3E} - s_{EH})}.$$

Thus, the starting time of control is:

$$t'_V = t_V + \Delta t_X - 3\Delta t = t_V + \frac{l_{IV} - l_V}{(s_{A_3E} - s_{EH})} - 3\Delta t.$$

15. Eq. (15): derived from the geometric relationship shown on Fig. 7(b).
Firstly, according to the geometry on Fig. 7(b), we have the following relationship:

$$\Delta t_{\widetilde{E}} v_H + w * (\Delta t_{\widetilde{a_{E^*}H}} - \Delta t_{\widetilde{E}}) = s_{\widetilde{a_{E^*}H}} * \Delta t_{\widetilde{a_{E^*}H}},$$

in which $\Delta t_{\widetilde{a_{E^*}H}}$ is the temporal duration of shock $s_{\widetilde{a_{E^*}H}}$.
Solving the equation above, we have:

$$\Delta t_{\widetilde{a_{E^*}H}} = \frac{\Delta t_{\widetilde{E}}(v_H - w)}{s_{\widetilde{a_{E^*}H}} - w}.$$

Notice that the length, l_4^R, is given by:

$$l_4^R = \left(v_{E^*} - s_{\widetilde{a_{E^*}H}}\right)\Delta t_{\widetilde{a_{E^*}H}} = \left(v_{E^*} - s_{\widetilde{a_{E^*}H}}\right)\frac{\Delta t_{\widetilde{E}}(v_H - w)}{s_{\widetilde{a_{E^*}H}} - w},$$

in which

$$v_{E^*} = \frac{w q_{E^*}}{(q_{E^*} + k_j w)},$$
$$s_{\widetilde{a_{E^*}H}} = \frac{q_{\widetilde{a_{E^*}}} - q_H}{k_{\widetilde{a_{E^*}}} - k_H},$$
$$k_{\widetilde{a_{E^*}}} = k_A,$$
$$q_{\widetilde{a_{E^*}}} = k_{\widetilde{a_{E^*}}} v_{E^*}.$$

16. Eq. (16): derived from the FD on Fig. 7(a).
17. Eq. (17): derived from the geometric relationship shown on Fig. 9(a).
18. Eq. (18): derived from the geometric relationship shown on Fig. 9(a).

A.1

Here we provide the simpler VSL control strategy in the case of overestimated q_{BN}. Fig. A-1(a) shows all traffic states on the FD and Fig. A-1(b) shows the traffic evolution.

Without additional VSL control upstream at t_V, state A_3 (zone III) interacts with state H (zone VI), forming a shock, s_{A_3H}. Zone III is eventually terminated when s_{A_3H} collides with shock $s_{A_2A_3}$. Similar transitions occur to A_2 (zone II) and A_1 (zone I) sequentially. In this process, the four transition zones (I-IV) are terminated by the queue in H gradually, and the speed transition from A to H becomes gradually more abrupt. Eventually, state H interacts with state A directly. The delay saving resulting from this strategy is the same as the more elaborate strategy in Section 3.3.

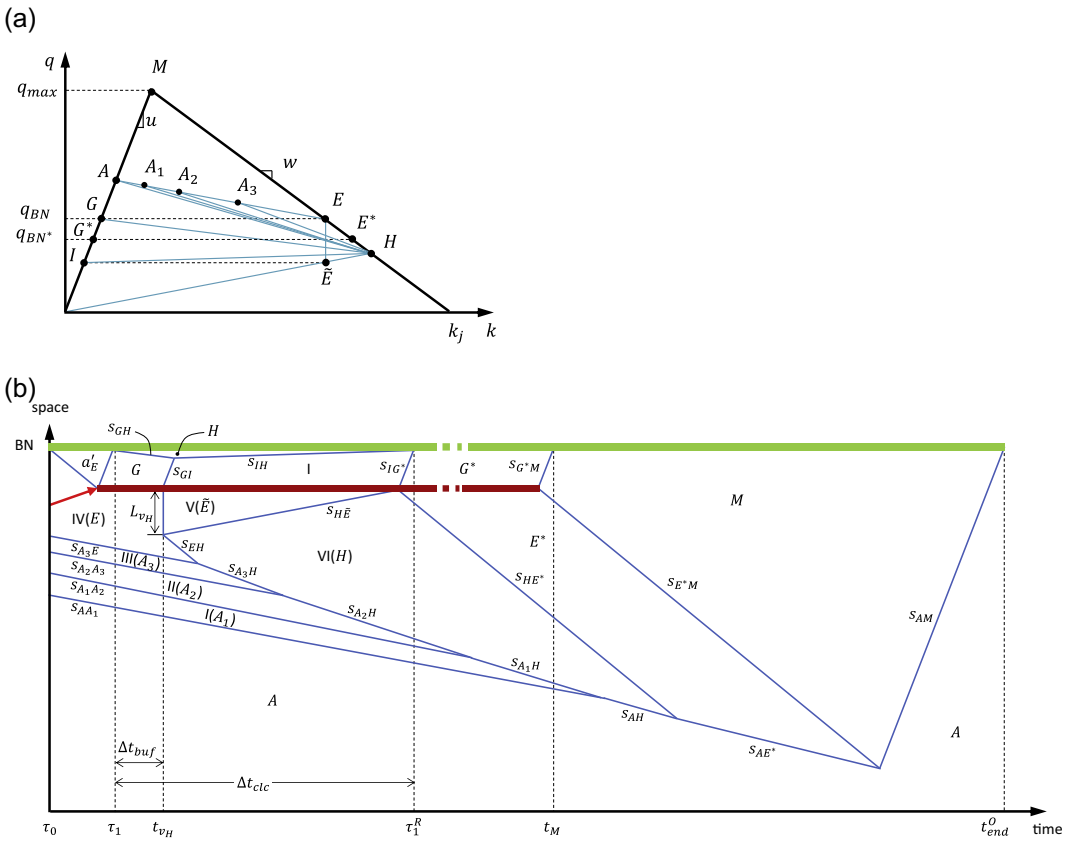


Fig. A-1. Simpler VSL control strategy to address overestimated q_{BN}: (a) Fundamental diagram; (b) Time–space diagram.

A.2

Here we discuss the extension of the VSL strategy in Section 3.3 to set $l_V = l_{IV}$ by using a lower speed limit, v_C. The FD and traffic evolution are shown in Fig. A-2. Based on the geometry shown in Fig. A-2(b), we can derive the required speed limit that results in $l_{IV} = l_V$. Since $l_{IV} < l_V$ is most likely to occur in late queue detection, here we assume that the queue is detected at the entrance of the acceleration zone, and the required speed limit, v_C, can be derived by solving the following equation:

$$\frac{l_{IV} - L}{v_C} + \frac{L}{u} = \frac{L}{s_{JH}},$$

in which state J has the flow $q_J = k_E v_C$, but free-flow speed, and thus $k_J = \frac{q_J}{u}$. Thus we have:

$$s_{JH} = \frac{q_H - q_J}{k_H - k_J},$$

Solving the first equation for v_C after plugging in s_{JH}, we obtain:

$$v_C = \frac{l_{IV} q_H u w}{(l_{IV} + L)u(q_{BN} + k_j w) - L(q_{BN} u - q_H(u - w))}.$$

With v_C we can obtain the length of zone V using the following relationship shown on Fig. A-2(b):

$$l_V = v_C\left(\Delta t_{clc} - \Delta t_{buf} - \frac{L}{u}\right).$$

A.3

Here we illustrate the case of VSL control strategy in Section 3.3 when the spatial extent of zone IV is not sufficiently long to dissipate the re-emergent queue; i.e., $l_{IV} > l_V$. For simplicity, we build on the simpler strategy described in Appendix A.1. Fig. A-3(a) and 3(b) show the FD and the traffic evolution, respectively. As shown in Fig. A-3(b), the temporal extent of zone V

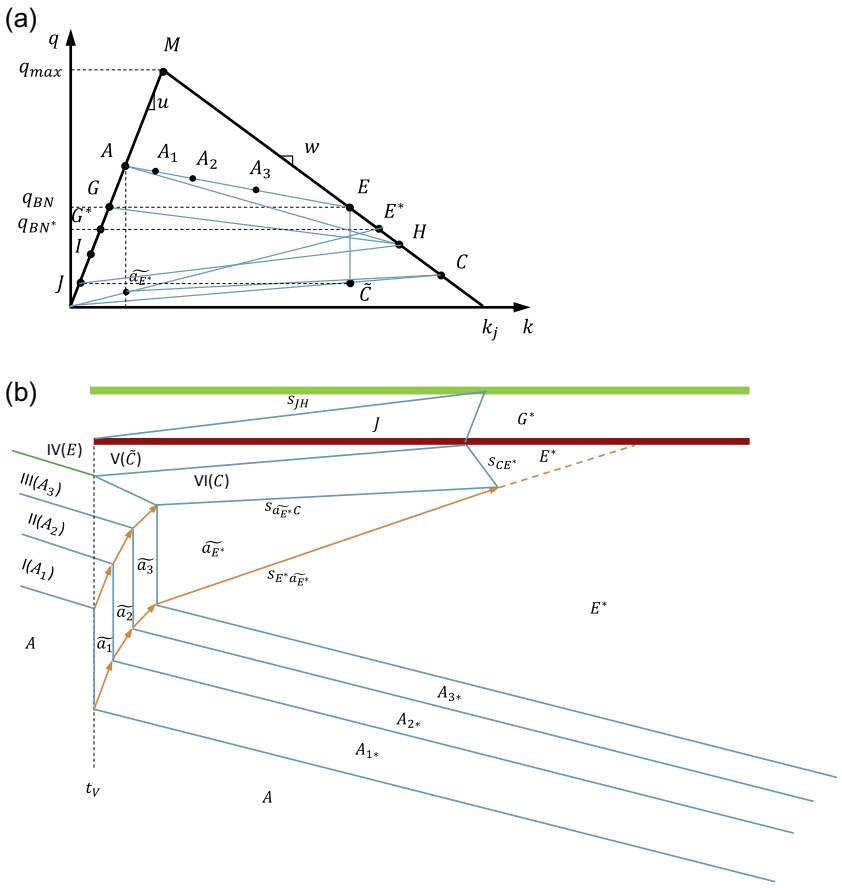


Fig. A-2. Simpler VSL control strategy to address overestimated q_{BN} when lower speed limit blow v_H is used: (a) Fundamental diagram illustrating traffic states that arise with VSL; (b) Time-space diagram illustrating shock waves generated by VSL; In (b), the traffic evolution before t_V is not shown but it is the same as Fig. 8(b).

(in $\widetilde{E}$) is not sufficient to dissipate queue in H. To address this problem, we extend the vertical control to zone III (in A_3), which results in state $\widetilde{A_3}$. Traffic emerging from $\widetilde{A_3}$ evolves to I_3 after passing the acceleration zone. The transition between I_3 and queue H forms a forward moving shock, s_{I_3H}. Note that this shock travels much faster than s_{IH} because state I_3 has much lower flow. It is likely that zone III is sufficient to dissipate the residual of queue H as shown by Fig. A-3(b) since the temporal extent of state I_3, Δt_{I_3}, is quite extensive:

$$\Delta t_{I_3} = \frac{l_{III}}{v_H},$$
$$l_{III} = (V_3 - w)\Delta t,$$

where l_{III} is the length of zone III.

In this case, we can resume high discharge flow q_{BN^*} after shock s_{I_3H} has arrived at the BN, similar to *Step O2*. The traffic evolution upstream of zone III is very similar to the evolution described in Appendix A.1; i.e., A_2 naturally evolves to state H and then E^*. So do A_1 and A. In case that zone III is not sufficient to resolve H, the vertical control can be extended further, even to state A. The traffic evolution is very similar. Notice that to decide the extent of vertical control, we need to check logical statements i-1, i-2, i-3, i-4 given below in sequence. If (i-1) is true, it means that the vertical control can be applied within zone IV. If not, we proceed to check whether (i-2) holds. Similar judgment applies. If (i-4) does not hold, it means that even the four zones (I to IV) are not sufficiently long, and we have to extend the vertical control to A. However, that's rare because the total duration of states I_3, I_2, and I_1 is quite long. Nevertheless, if that occurs, a similar action can be applied.

$$L < s_{IH}\Delta t_I, \tag{i-1}$$
$$L < s_{IH}\Delta t_I + s_{I_3H}\Delta t_{I_3}, \tag{i-2}$$
$$L < s_{IH}\Delta t_I + s_{I_3H}\Delta t_{I_3} + s_{I_2H}\Delta t_{I_2}, \tag{i-3}$$
$$L < s_{IH}\Delta t_I + s_{I_3H}\Delta t_{I_3} + s_{I_2H}\Delta t_{I_2} + s_{I_1H}\Delta t_{I_1}. \tag{i-4}$$

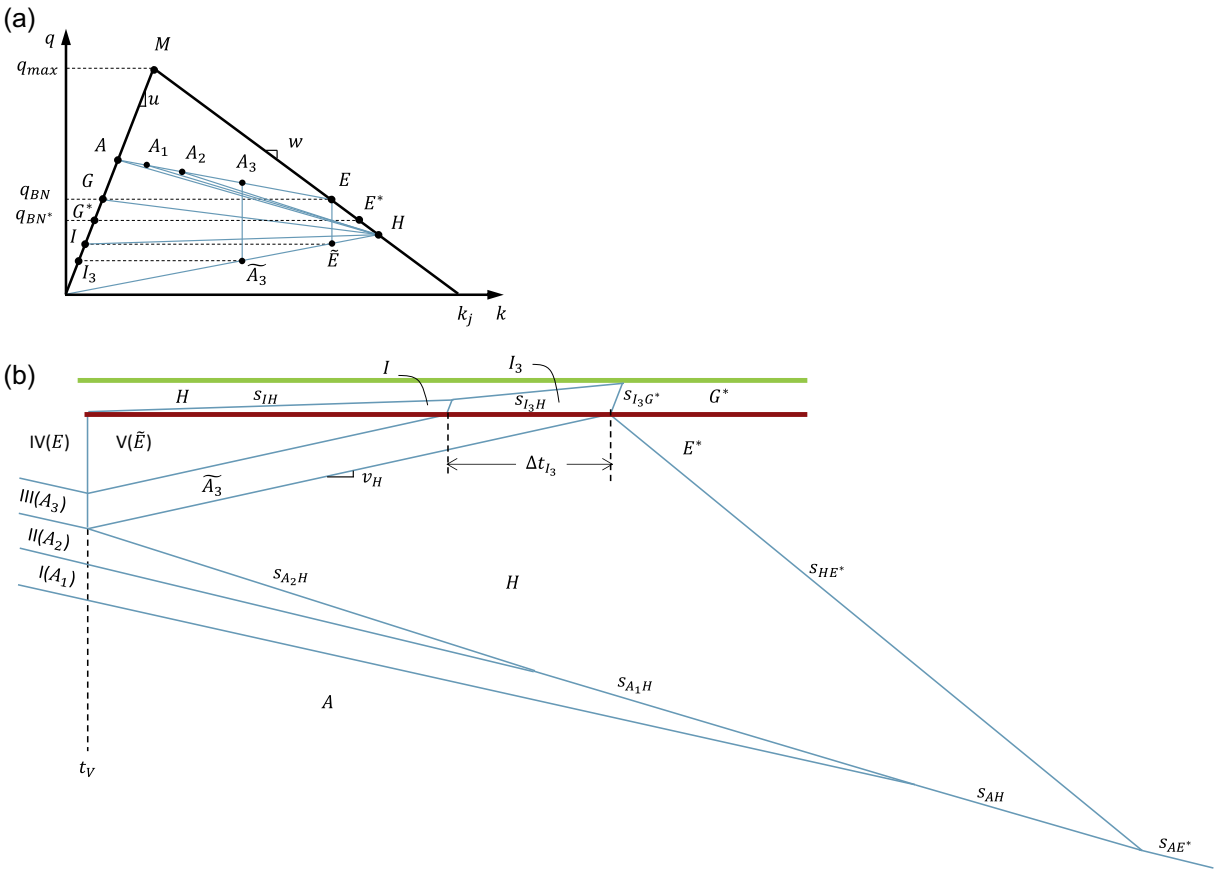


Fig. A-3. Simpler VSL control strategy to address overestimated q_{BN} when zone IV is not sufficient to resolve re-emergent queue: (a) Fundamental diagram illustrating traffic states that arise with VSL; (b) Time–space diagram illustrating shock waves generated by VSL when vertical control is extended to zone III; In Fig. A-3(b), the traffic evolution before t_V is not shown but it is the same as Fig. 5(b).

Notably, it is possible that only a partial zone is needed. In this case, we can back calculate the needed duration of states I_3, I_2, and/or I_1. For instance, for the case of vertical control partially extending to zone III, the required duration of I_3, $\Delta t_{I_3'}$, is:

$$\Delta t_{I_3'} = \frac{L - s_{IH}\Delta t_I}{s_{I_3H}}. \tag{i-5}$$

And the spatial extent needed in zone III, $l_{I_3'}$, is:

$$l_{I_3'} = \Delta t_{I_3'}\, v_H. \tag{i-6}$$

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